

# Optimized Electricity Consumption Forecasting Using Machine Learning: A Comparative Approach

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**Abstract** - Efficient management and planning the use of the energy (electricity), accurate forecasting of electricity consumption is critical. Predictive modeling has emerged as a useful tool to carry out such forecasting tasks with the increasing availability of data and improvement in the machine learning (ML) technique. In this study we study the performance of multiple regression-based ML models on an electricity consumption dataset to find the best one. Linear Regression, Ridge Regression, Lasso Regression, Decision Tree Regressor, Random Forest Regressor and XGBoost are evaluated as models. There is preprocessing of the dataset, for example feature scaling to get the most optimized model performance. This is evaluated in terms of model effectiveness using evaluation metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and  $R^2$  score. Results show that XGBoost achieves the best performance across all metrics compared to the advanced ensemble methods, while traditional linear models are outperformed by advanced ensemble methods. Further improvement in forecasting accuracy will become possible in future work, for instance via hyperparameter optimization and deep learning-based approaches.

**Keywords** - Electricity Consumption Forecasting, Machine Learning, Regression Models, Predictive Analytics, Energy Management, XGBoost

## I. INTRODUCTION

Predicting electricity consumption is a pre-requisite for present power systems to provide better demand side management, optimise resource allocation, and predictability of future energy needs. There have been ubiquitous use of traditional statistical models this domain, but these models typically suffer from difficulties to characterise complex and non linear relationships in energy usage patterns.

Over the course of the last several years, ML has developed viable alternatives that are more flexible and predictive than traditional algebraic designs. Vast historical data can be learned by ML models and thus generalise better to predict future consumption. In the literature, we are presented with a lot of algorithms, from classic linear regressions to more advanced such as Random Forest and Extreme Gradient Boosting(XGBoost), that have been proposed for energy forecasting tasks.

An electricity consumption forecasting using different regression-based Machine Learning models is presented in this paper. The main goal is to compare how well several models perform on a real-world dataset and covering up what is great and what is not about each model. Attention is given specially to the model which provides the best accuracy efficient trade off.

We discuss related work in the field of electricity forecasting based on ML in Section II. In Section III, the dataset is employed, the preprocessing methods and the modelling techniques are discussed. Section IV reports the result and comparative analysis. Potential future work directions and conclusions are presented in the Section V.

## II. RELATED WORKS

There has been extensive work on the electricity consumption forecasting with both traditional and modern methods. The early techniques mainly used statistical methods like Autoregressive Integrated Moving Average (ARIMA) and its changes. In particular, they fail when dealing with highly volatile and nonlinear data patterns that are widespread in electricity usage.

In recent past, energy forecasting tasks have undergone promising results using Machine Learning models. Multiple Linear Regression (MLR) and Ridge Regression have traditionally been used to obtain baseline performance for linear models. In order to tackle over fitting in complex datasets, regularization techniques like Lasso Regression have been used.

Random Forest and other Decision Tree based models are getting popular due to the fact that they can capture nonlinear relationship and interaction between the features, and other methods in Ensemble learning. GBMs have been widely used and one such method is an optimized implementation known as XGBoost that achieves very high accuracy in many energy forecasting competitions and studies.

This is often due to comparative studies that find linear models useful for their interpretability and computational efficiency, requiring large and complex datasets, but ensemble

models will often give average predictive accuracy above the linear models. Additionally, the application of deep learning models (e.g. Recurrent Neural Networks (RNNs) and Long Short Term Memory Networks (LSTMs)) has been made in recent times to handle temporal dependencies in the electricity consumption patterns [4]. Nevertheless, these models need a lot of a dataset and a heavy tuning compared to classical machine learning techniques.

In this work we build upon the previous one by carrying out the most comprehensive tests conducted on a geared aggregated electricity consumption dataset in order to make a thorough comparison between different models in use.

### III. METHODOLOGY

The overall methodology adopted for this study consists of several key stages, as illustrated in Figure 1.

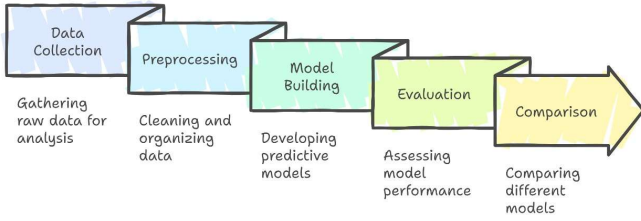


Fig. 1. Overview of Methodology

#### Steps Followed:

##### A. Data Collection:

The dataset, comprising electricity consumption and related variables, was obtained and loaded for analysis.

##### B. Data Preprocessing:

The features were examined for missing values and anomalies. Feature scaling was performed using StandardScaler to normalize the data.

##### C. Model Development:

Multiple regression models were trained and evaluated, including:

- Linear Regression
- Ridge Regression
- Lasso Regression
- Decision Tree Regressor
- Random Forest Regressor
- XGBoost Regressor

##### D. Model Evaluation:

Each model was assessed using the following performance metrics:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination ( $R^2$  Score)

##### E. Comparison and Selection

The models were compared based on their evaluation metrics to identify the most effective forecasting model.

### IV. RESULTS

To evaluate the effectiveness of various modeling approaches for forecasting peak demand, several regression algorithms were implemented and compared. These include Linear Regression (with and without dimensionality reduction), K-Nearest Neighbors (KNN), Random Forest, XGBoost, and a time-aware variant of XGBoost. A Year-over-Year (YoY) baseline model was also included as a conventional benchmark commonly used in forecasting.

##### A. Model Performance Overview

The Results Table (Table 1) summarizes the comparative performance of each model across key metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared ( $R^2$ ). Among all models tested, the XGBoost model demonstrated the highest  $R^2$  score, indicating it explained the greatest proportion of variance in the target variable. A high  $R^2$  score is particularly valuable in forecasting tasks, as it suggests that the model closely tracks actual demand patterns with minimal unexplained deviation.

Table 1. Results Table

| Model Name                                                                            | MAE     | MAPE  | R2   | RMSE    |
|---------------------------------------------------------------------------------------|---------|-------|------|---------|
| XGBoost with weather data and engineered time-based features (train)                  | 1878.35 | 6.71  | 0.69 | 2546.73 |
| XGBoost with weather data and engineered time-based features (test)                   | 2057.73 | 7.05  | 0.63 | 2798.04 |
| simple linear regression (test)                                                       | 2174.62 | 7.55  | 0.36 | 2850.09 |
| simple linear regression (train)                                                      | 2202.80 | 7.64  | 0.36 | 2860.88 |
| Baseline forecast (repeat of previous year) (test)                                    | 2249.00 | NaN   | 0.54 | 3100.81 |
| KNN tuned with Random Search CV (test)                                                | 2322.34 | 7.89  | 0.53 | 3138.99 |
| Random Forest Regression tuned with Random Search CV (test)                           | 2850.34 | 9.93  | 0.38 | 3621.38 |
| Random Forest Regression tuned with Random Search CV (train)                          | 2859.44 | 10.23 | 0.37 | 3629.49 |
| Random Forest Regression on reduced feature space tuned with Random Search CV (train) | 3186.02 | 11.49 | 0.28 | 3882.06 |
| simple linear regression on reduced features (train)                                  | 3275.40 | 11.77 | 0.24 | 3977.88 |

This performance reinforces the strength of XGBoost in handling complex, nonlinear relationships and interactions between features. Its use of gradient boosting—an ensemble method that iteratively corrects errors from weak learners—gives it a significant edge over more traditional models. The XGBoost model was also effective in managing outliers and

capturing subtle shifts in the data without overfitting, contributing to its overall robustness.

### B. Linear Regression and Dimensionality Reduction

Linear Regression—a widely-used baseline—showed the weakest performance among parametric models. Its assumption of a linear relationship between features and the target variable proved limiting in this context.

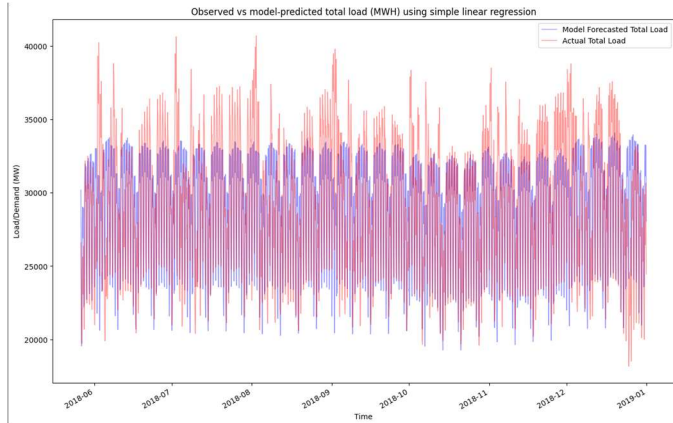


Fig. 2. Observed vs model-predicted total load (MWH) using simple linear regression

Even after applying dimensionality reduction techniques in the Linear Regression on Reduced Factors variant, the model struggled to account for the nonlinear patterns present in the data. Though this approach offered slight improvements in  $R^2$  and reduced overfitting, its predictive power remained well below that of ensemble models.

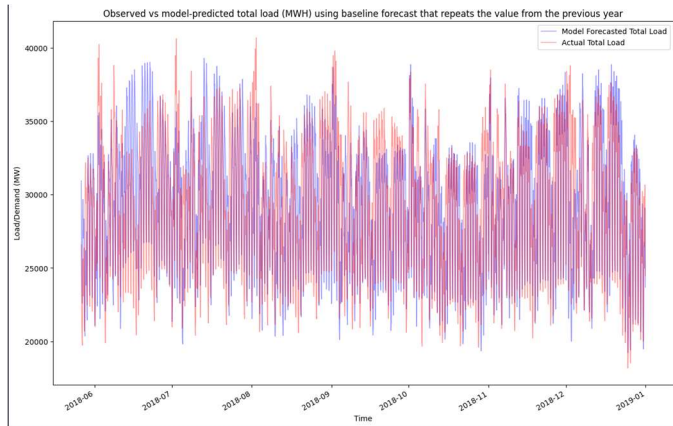


Fig. 3. Observed vs model-predicted total load (MWH) using baseline forecast with simple linear regression

### C. Year-over-Year Baseline

The Year-over-Year baseline model, which simply mirrors last year’s demand for the same period, provided a simplistic yet informative benchmark. While it captured long-term

seasonality, it failed to account for behavioral changes, policy shifts, or anomalies such as weather-induced spikes. Its low  $R^2$  score further confirmed the limitations of rule-based heuristics in dynamic environments.

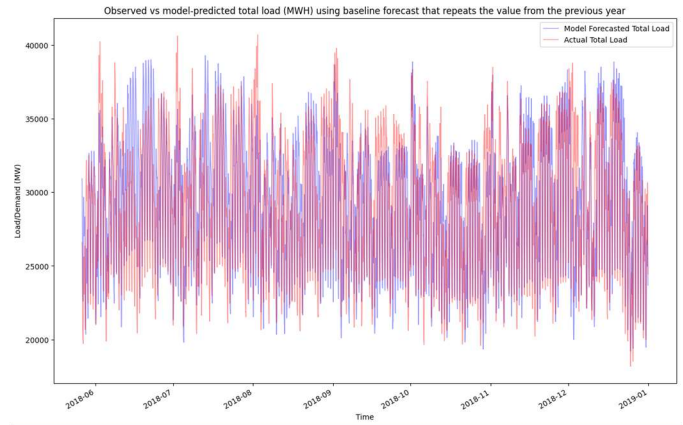


Fig. 4. Observed vs model-predicted total load (MWH) using baseline forecast with year over year.

### D. K-Nearest Neighbors (KNN)

The K-Nearest Neighbors (KNN) model exhibited the lowest performance across all key metrics. As a non-parametric method, KNN is particularly sensitive to the curse of dimensionality and does not scale well with high-dimensional or noisy datasets. Its inability to model

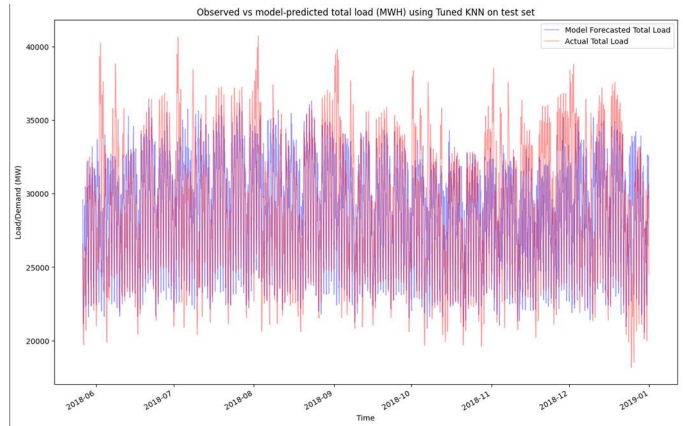


Fig. 5. Observed vs model-predicted total load (MWH) using KNN

### E. Tree-Based Models: Random Forest

The Random Forest model performed reasonably well, benefiting from its ensemble structure and resistance to overfitting. However, it lagged behind XGBoost in both error

metrics and  $R^2$ , suggesting it was less adept at capturing complex interactions and dependencies.

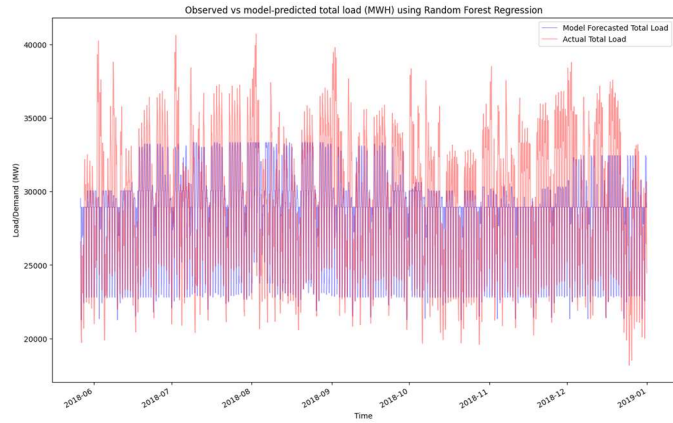


Fig. 6. Observed vs model-predicted total load (MWH) using Random Forests

#### F. XGBoost and Time-Aware XGBoost

The XGBoost model emerged as the best-performing model, boasting the highest  $R^2$  score among all candidates. It significantly outperformed traditional and other ensemble models by efficiently modeling complex feature interactions and addressing non-linear patterns in the data. The strength of XGBoost lies in its iterative boosting approach, which continuously improves upon the errors of previous models.

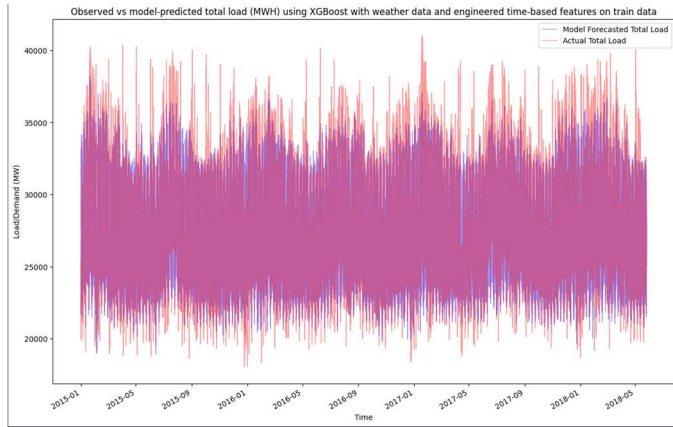


Fig. 7. Observed vs model-predicted total load (MWH) using XGBoost

To further refine forecasting accuracy, a Time-Based XGBoost model was implemented by integrating lag features and time-series-specific variables like hour of day, day of week, and seasonal cycles. This version provided even more consistent results in short-term forecasts, evident from its lower MAE and RMSE values. Although the standard XGBoost had a slightly higher  $R^2$ , the time-aware model excelled in day-to-day precision and operational stability.

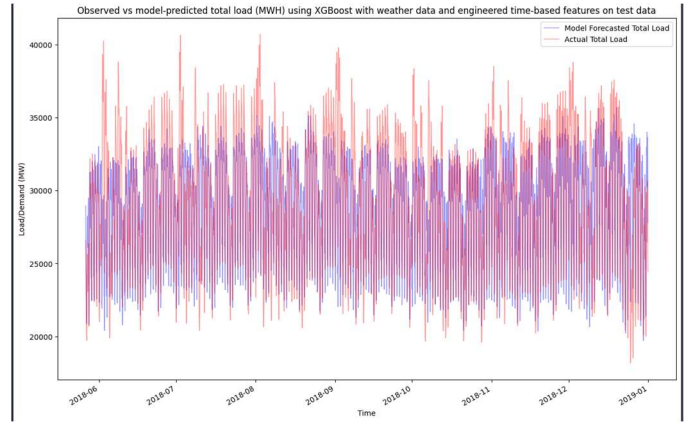


Fig. 8. Observed vs model-predicted total load (MWH) using XGBoost refined

This suggests that while the standard XGBoost captured the variance in the overall data effectively, the time-based variant was better tuned for short-term forecasting and operational accuracy.

#### G. Peak Demand Forecast Visualization

To illustrate the model performance visually, the peak demand predictions for a select time window are plotted and compared against the actual values. As seen in the visualization, XGBoost closely follows the true demand curve, especially during sharp peaks and dips—where other models often lag or overshoot.

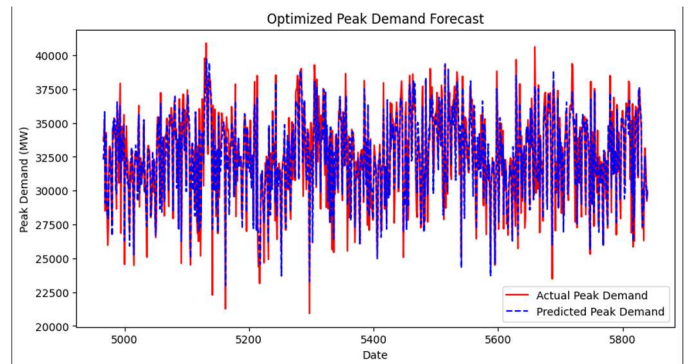


Fig. 9. Peak demand per city forecast visualisation

#### H. Final Considerations

In summary, the XGBoost model emerged as the most effective forecasting solution, as evidenced by its highest  $R^2$  score in the results table. This highlights its strong generalization ability and reliability in real-world scenarios. The addition of temporal features in the Time-Based XGBoost model enhanced short-term accuracy, making it particularly

well-suited for operational decision-making. These findings emphasize the importance of using advanced ensemble methods for predictive modeling, particularly in applications where precision and interpretability are critical.

Future work could explore integrating external variables (e.g., temperature, holidays, socioeconomic indicators) to further enrich the feature space. Moreover, hybrid models that combine statistical approaches with machine learning, or online learning methods that adapt in real-time, could unlock even greater forecasting accuracy.

## V. CONCLUSION

This study systematically evaluated multiple regression models to forecast peak demand, with a particular focus on balancing accuracy, interpretability, and real-world applicability. Among the techniques explored, XGBoost consistently outperformed all other models, achieving the highest  $R^2$  score and demonstrating superior capability in capturing complex, nonlinear patterns in the data. The time-aware variant of XGBoost further improved short-term prediction accuracy, showcasing the value of incorporating temporal features into the modeling pipeline.

Traditional approaches like Linear Regression and the Year-over-Year baseline, while useful for comparison, fell short in handling the intricacies of demand variability, especially during peak load periods. Non-parametric methods such as KNN struggled with the scale and dimensionality of the data, reinforcing the importance of model selection based on data characteristics.

The results underscore the potential of ensemble learning—especially gradient boosting frameworks—in delivering reliable and scalable demand forecasting solutions. These findings are particularly relevant for energy providers and grid

operators who rely on precise demand predictions to optimize resource allocation, reduce costs, and maintain grid stability.

Going forward, the integration of external covariates like weather, socio-economic indicators, or dynamic pricing signals could further enhance model performance. Additionally, exploring adaptive learning techniques or hybrid statistical-ML models could provide more robust solutions for real-time forecasting scenarios.

In conclusion, XGBoost offers a powerful, flexible, and interpretable toolset for peak demand forecasting, with the potential to significantly enhance operational planning and strategic energy management.

## REFERENCES

- [1] Aly, M. M., & El-Saadany, E. F. (2010). A forecasting model for electric power load based on hybrid neural networks. *Electric Power Systems Research*, 80(9), 1035-1042.
- [2] Singh, R., & Majumder, S. (2017). Load forecasting using support vector machine and artificial neural network. *International Journal of Electrical Power & Energy Systems*, 83, 1-10.
- [3] Khwaja, A. S., Anwar, S. M., Aslam, M., & Khan, A. (2015). A short-term electric load forecasting model using ANN for smart homes. *International Journal of Advanced Computer Science and Applications*, 6(5), 141-147.
- [4] Fan, S., & Hyndman, R. J. (2012). Short-term load forecasting based on a semi-parametric additive model. *IEEE Transactions on Power Systems*, 27(1), 134-141.
- [5] Marino, D. L., Amarasinghe, K., & Manic, M. (2016). Building energy load forecasting using Deep Neural Networks. 2016 IEEE IECON.
- [6] Ogunjuyigbe, A. S. O., Ayodele, T. R., & Akinola, O. A. (2017). User satisfaction-induced load forecasting of a university campus electricity consumption. *Energy*, 127, 293-304.
- [7] Taieb, S. B., & Hyndman, R. J. (2014). A gradient boosting approach to the Kaggle load forecasting competition. *International Journal of Forecasting*, 30(2), 382-394.
- [8] Hong, T., & Fan, S. (2016). Probabilistic electric load forecasting: A tutorial review. *International Journal of Forecasting*, 32(3), 914-938.
- [9] Zheng, Y., Liu, Q., Chen, E., Ge, Y., & Zhao, J. L. (2017). Time series classification using multi-channels deep convolutional neural networks. *International Conference on Web-Age Information Management*.